Microsoft Elevate

CAPSTONE PROJECT

Global Data Jobs Market Intelligence Dashboard

PRESENTED BY

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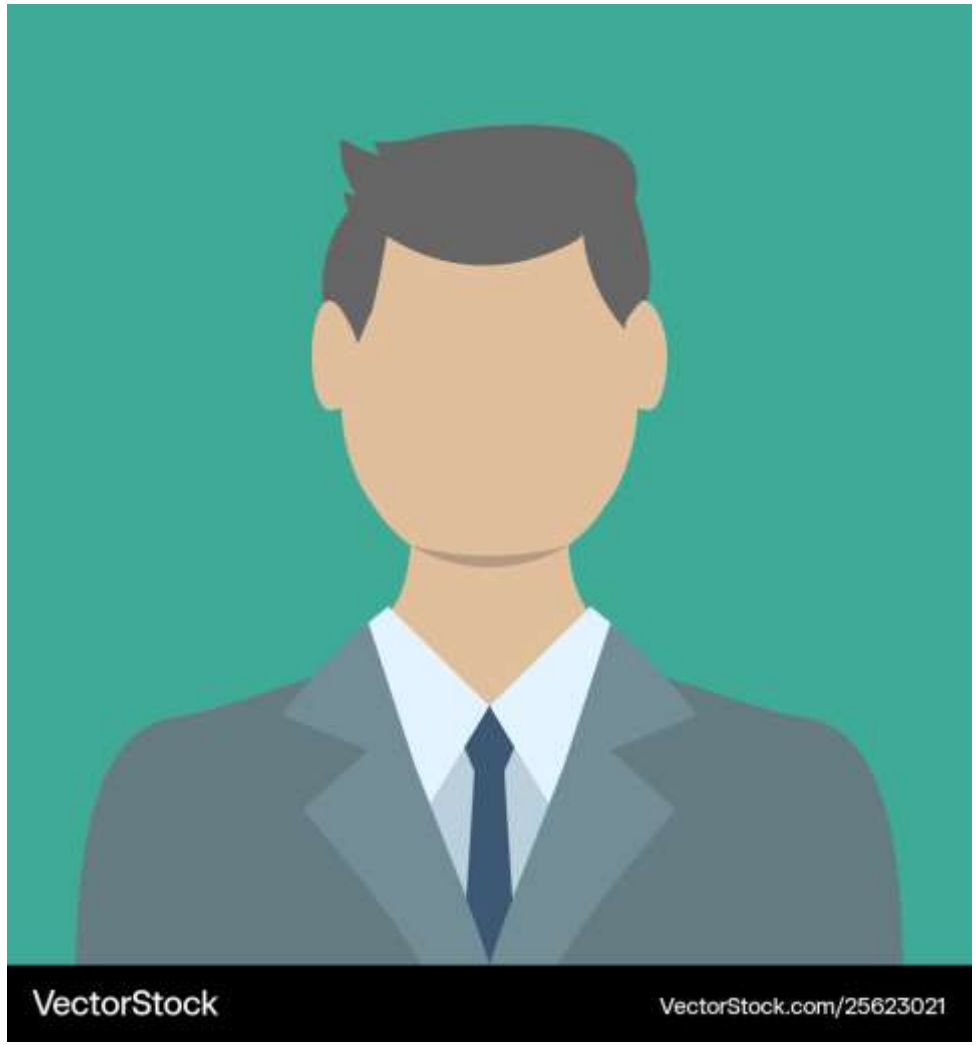

OUTLINE:

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PROBLEM STATEMENT:

- The rapid growth of data-driven industries has increased the demand for roles such as Data Analysts, Data Engineers, Data Scientists, and Business Analysts. However, job seekers often lack clear insights into job market trends, required skills, salary ranges, and job availability across different locations.
- Moreover, analyzing trends like remote job opportunities, global job availability, and required qualifications is challenging because job information is distributed across multiple platforms.
- Without a centralized system to analyze and visualize this information, it becomes difficult for individuals to make informed career decisions in the data job market.

Therefore, there is a need for a system that can analyze job postings and provide clear insights into job demand, salary trends, and employment opportunities in the data industry.

PROPOSED SOLUTION:

- The proposed system is an interactive Data Jobs Market Intelligence Dashboard developed using Power BI. It analyzes job posting data and converts raw information into clear visual insights about the global data job market.
- The dashboard helps users understand job demand, salary trends, required skills, and employment opportunities through interactive visualizations and filters.

Key Features

✓ **Job Market Trends** – Shows monthly trends in job postings to analyze demand changes over time.

✓ **Salary Analysis** – Compares hourly and yearly salaries across roles.

✓ **Job Role Demand** – Displays the number of job postings available for different job titles.

✓ **Remote Work Opportunities** – Highlights the percentage of work-from-home jobs.

✓ **Job Platforms** – Identifies major platforms where jobs are posted.

✓ **Job Schedule Types** – Analyzes job types such as full-time, contract, or internship roles.

✓ **Global Job Distribution** – Shows job availability across countries using a map visualization.

SYSTEM APPROACH:

- The system approach outlines the methodology used to develop the Global Data Jobs Market Intelligence Dashboard using Power BI.

The system follows these approach:

- ✓ **Data Collection** – Job posting data including job titles, salaries, locations, skills, and company information was collected from public datasets.
- ✓ **Data Preprocessing** – The dataset was cleaned and transformed using Power Query by handling missing values and formatting the data
- ✓ **Data Modeling** – Tables were structured and relationships were established in Power BI for efficient data analysis.
- ✓ **Data Visualization** – Charts such as line charts, bar charts, maps, scatter plots, and KPI cards were used to present insights.
- ✓ **Interactive Dashboard** – Filters and drill-through features allow users to explore detailed insights for specific job roles.

SYSTEM REQUIREMENTS:

Software Requirements

- Microsoft Power BI Desktop
- Data Jobs Dataset (Kaggle)
- Microsoft Excel– Used to view or inspect the dataset
- Operating System: Windows 10 / Windows 11

Hardware Requirements

- Computer or Laptop
- Minimum 8 GB RAM (recommended for Power BI performance)
- Storage: At least 10 GB free disk space
- Internet Connection

Libraries / Tools Used:

- **Power BI Desktop** – Used for building dashboards and visualizations
- **Power Query** – Used for data cleaning and transformation
- **Data Modeling** – Used to establish relationships between tables
- **DAX (Data Analysis Expressions)** – Used for creating calculated measures and metrics
- **Interactive Visualizations** – Charts, maps, KPI cards, and filters used in the dashboard

ALGORITHM & DEPLOYMENT:

This project uses Business Intelligence and Data Analytics techniques instead of machine learning algorithms.

The analysis is performed using:

✓ Data aggregation

✓ Data filtering

✓ Data transformation

✓ DAX calculations in Power BI

These methods help convert raw job posting data into meaningful insights.

Data Input

The dataset contains job posting information such as:

- Job title
- Company name
- Job location and country
- Job posting date
- Work-from-home availability
- Job schedule type (Full-time, Contract, etc.)
- Salary information (hourly and yearly)
- Job platform (LinkedIn, Indeed, etc.)

This data is imported into Power BI for analysis.

ALGORITHM & DEPLOYMENT:

Data Processing

The dataset is processed using Power Query with the following steps:

- Removing unnecessary columns
- Handling missing values
- Formatting salary fields
- Creating calculated measures using DAX

These steps ensure the dataset is clean and structured for analysis.

Deployment

The final solution is deployed as an interactive Power BI dashboard. Users can:

- Analyze job market trends over time
- Compare salaries across job roles
- Identify high-demand data jobs
- Explore global job opportunities
- Analyze remote work availability

Users interact with the dashboard using filters and drill-through features.

RESULT:

The developed dashboard presents key insights about the data job market through interactive visualizations.

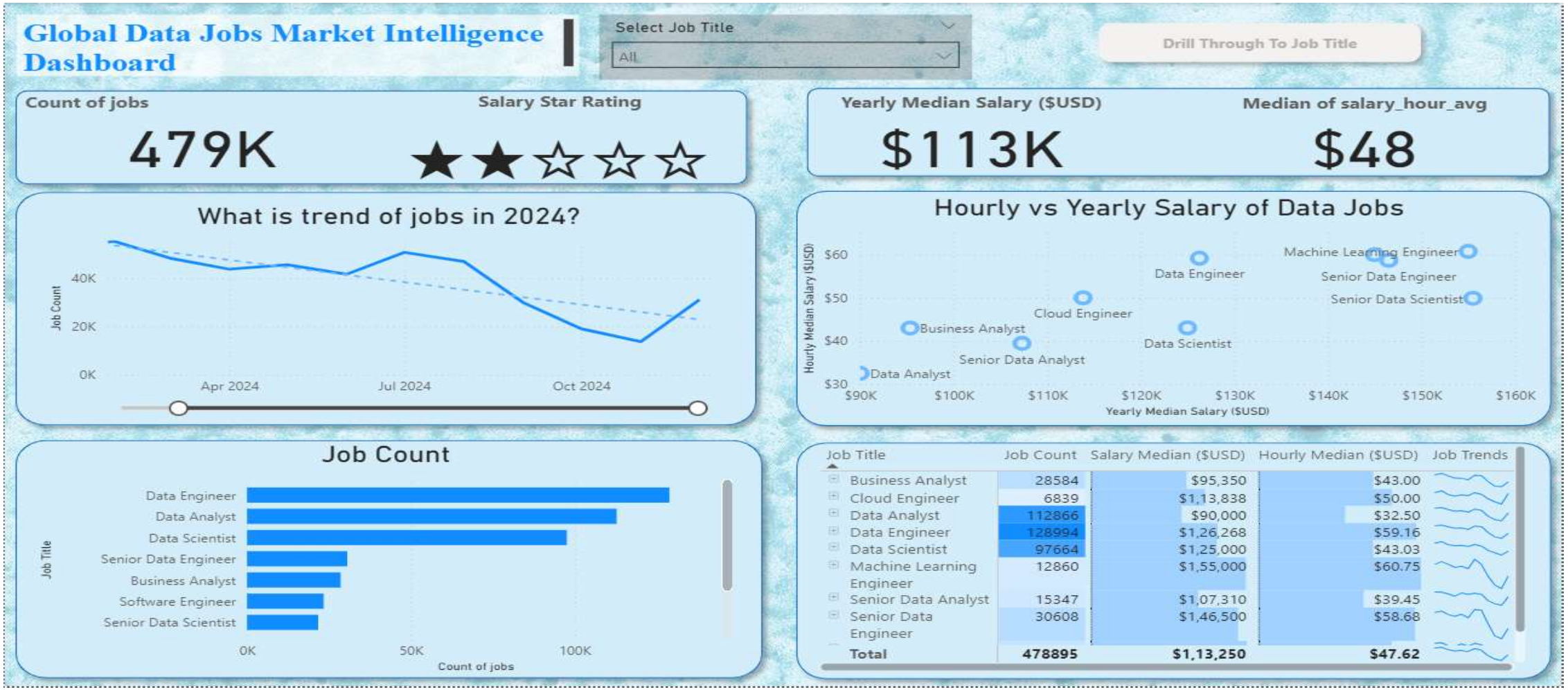
Key insights include:

- ✓ Identification of the most in-demand job roles
- ✓ Analysis of salary trends across different data jobs
- ✓ Visualization of job trends over time
- ✓ Analysis of remote work opportunities
- ✓ Global distribution of job postings

The interactive dashboard allows users to filter and explore insights based on job title and other parameters.

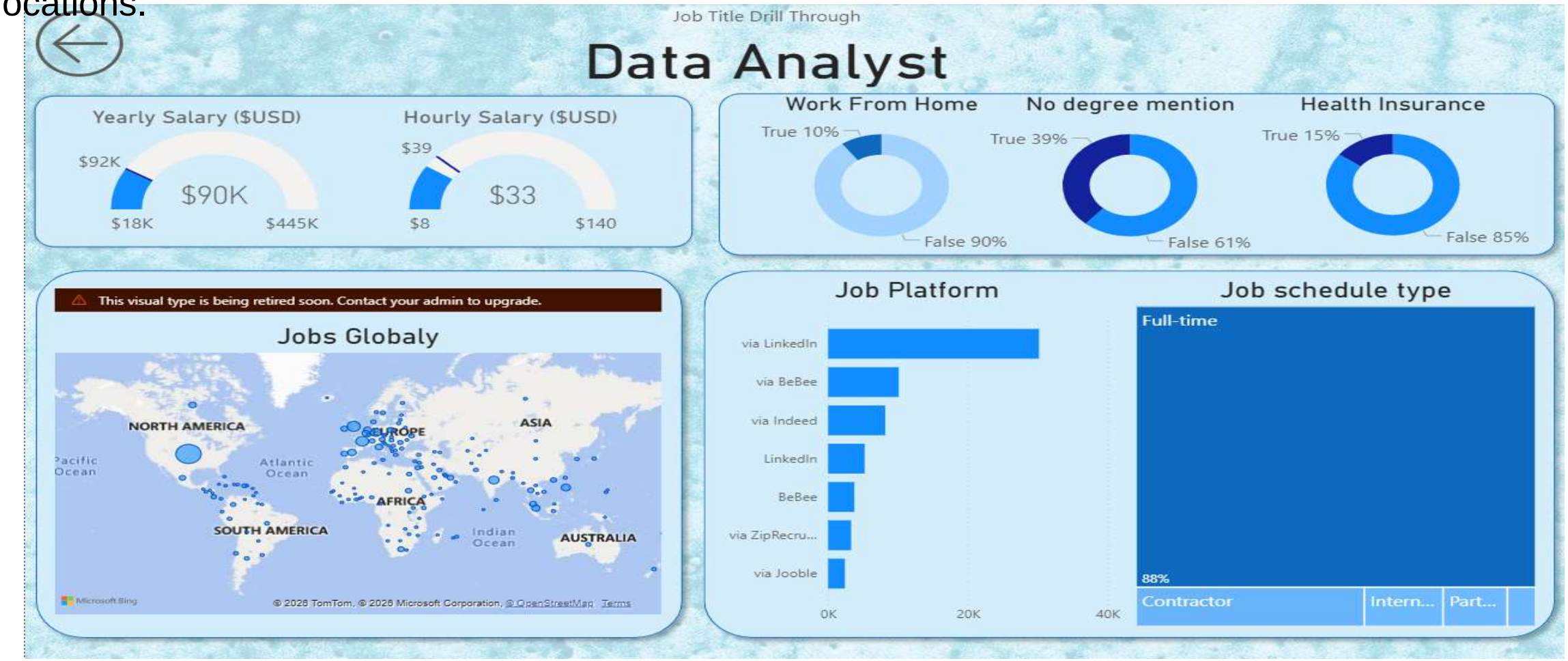
Dashboard Overview

The image displays the main dashboard showing job count, salary metrics, job trends, salary comparison across roles, and role-wise job distribution.



Job Role Drill-Through Analysis

The image shows a detailed drill-through view for a selected job role, presenting salary gauges, work-from-home statistics, job platform distribution, job schedule types, and global job locations.



CONCLUSION:

- The Global Data Jobs Market Intelligence Dashboard effectively analyzes job posting data and provides valuable insights into the data job market. The dashboard highlights job demand trends, salary distribution, remote work opportunities, and job availability across different roles and locations.
- The interactive visualizations enable users to easily explore job market patterns and compare key metrics. The use of filters and drill-through features allows users to perform deeper analysis for specific job roles.
- Overall, the dashboard demonstrates how Power BI can transform raw job data into meaningful insights for better understanding of the data job market.

FUTURE SCOPE:

The current dashboard provides valuable insights into job market trends in the data industry. However, the system can be further enhanced with several improvements.

- **Real-Time Data Integration** – Connect the dashboard with live job portals to analyze real-time job market trends.
- **Regional Expansion** – Extend the system to analyze job opportunities across multiple countries, cities, or regions for more detailed geographic insights.
- **Integration of Additional Data Sources** – Incorporate data from more job platforms such as LinkedIn, Glassdoor, and Indeed to provide a more comprehensive analysis of the job market.
- **Online Dashboard Deployment** – Publish the dashboard on Power BI Service to allow users to access the job market insights from anywhere.

These improvements can make the system more powerful and useful for job seekers, students, and industry professionals.

REFERENCES:

Data Jobs Dataset : https://www.kaggle.com/datasets/arshkon/linkedin-job-postings

Microsoft Power BI Documentation: https://learn.microsoft.com/en-gb/power-bi/

Power BI Official Website: https://www.microsoft.com/en-gb/power-platform/products/power-bi/

Project Source Link:
https://drive.google.com/drive/folders/1Ngk1hbDaVkyL8pNq3vs2ypdNtIbcVqCZ?usp=sharing

Thank You